

Aristide Tanyi-Jong Akem

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EDUCATION

PhD - Telematics Engineering Mar 2021 - Sep 2024

Universidad Carlos III de Madrid (UC3M), Madrid, Spain

Advisor: Marco Fiore

Thesis title: User-plane Algorithms for Stateless and Stateful Inference in Programmable Networks

Thesis committee: Albert Banchs, Gianni Antichi, and Salvatore Pontarelli

MSc - Electrical and Computer Engineering Aug 2019 - May 2020

Carnegie Mellon University Africa (CMU-Africa), Kigali, Rwanda

Advisor: Edwin Mugume

MEng - Telecommunications Engineering Sep 2013 - July 2018

University of Yaounde I (UYI), Yaounde, Cameroon

Advisor: Lele Chrislin

RESEARCH EXPERIENCE

University of Southampton (Southampton, United Kingdom) | Lecturer in Computer Science Sep 2025 - Present

Member of the Cyberphysical Systems Group in the School of Electronics and Computer Science, on a balanced pathway.

University of Oxford (Oxford, United Kingdom) | Visiting Researcher April 2026 - Present

Academic Visitor in the Department of Engineering Science.

University of Oxford (Oxford, United Kingdom) | Postdoctoral Researcher Feb 2025 - March 2026

Postdoc in the Computing Infrastructure Group led by Prof. Noa Zilberman, in the Department of Engineering Science.

IMDEA Networks Institute (Madrid, Spain) | Research Assistant Mar 2021 - Jan 2025

Carried out my PhD studies as an MSCA-ITN Fellow within the Networks Data Science Group led by Prof. Marco Fiore.

My research focused on user-plane inference with key results presented at conferences like IEEE INFOCOM.

Ranplan Wireless (Cambridge, United Kingdom) | Visiting Researcher Sep 2023 - Apr 2024

I visited Ranplan Wireless during the same period I was at the University of Cambridge, as part of the secondments of the BANYAN project. I also had a previous visit to Ranplan between October 2022 and February 2023.

University of Cambridge (Cambridge, United Kingdom) | Visiting PhD Student Sep 2023 - Feb 2024

I visited the Department of Computer Science and Technology under the supervision of Dr. Ian Wassell, as part of the secondments of my PhD program.

Orange Innovation Networks (Paris, France) | Visiting PhD Researcher Mar 2023 - Aug 2023

I visited the SMART team at Orange Innovation under Dr. Guillaume Fraysse, where we worked on encrypted traffic classification at line rate in programmable switches.

TEACHING EXPERIENCE

University of Oxford (Oxford, United Kingdom) | Laboratory Demonstrator June 2025

Course: Programmable Networks

University of Cambridge (Cambridge, United Kingdom) | Laboratory Assistant Oct 2023 - Feb 2024

Course: Hardware Laboratory

Carnegie Mellon University Africa (Kigali, Rwanda) | Graduate Teaching Assistant Aug 2020 - Mar 2021

Courses: Data Inference and Applied Machine Learning, Data Analytics and Big Data Science.

OMEGA Learning Centre (Yaounde, Cameroon) | Mathematics Instructor Sep 2018 - May 2019

Course: Mathematics

INDUSTRY EXPERIENCE

Ministry of ICT and Innovation (Kigali, Rwanda) | Data Science Intern May 2020 - Aug 2020

HW Technologies Co. Ltd. (Yaounde, Cameroon) | Intelligent Video Surveillance Engineer Jan 2019 - Jul 2019

INVITED TALKS

- Research Journey: From Problem to Scholarly Publication - *Ruzivo Research Lab, African Leadership University*, 27/02/2026
- In-Network Inference with P4: From Stateless to Hybrid Approaches - *P4 Developer Days*, 21/01/2026
- A Unified Model for Packet and Flow Inference in Programmable Networks - *Coseners 2025*, 04/07/2025
- Advancing Network Intelligence with Programmability and Machine Learning - *Carnegie Mellon University Africa*, 17/02/2025
- On-Device Machine Learning for Traffic Classification in Programmable Networks - *UPANZI Network Webinar Series*, 29/01/2025
- Stateless and Stateful User-Plane Algorithms for Inference in Programmable Networks - *Telefonica Research*, 28/08/2024
- Stateless and Stateful User-Plane Algorithms for Inference in Programmable Networks - *Purdue University*, 07/08/2024
- Towards High-Speed Network Intelligence with On-Device Machine Learning - *IBM Research Africa*, 25/07/2024

OTHER TALKS

- Towards Ultra-Low Latency User-Plane Cyberattack Detection in SDN-based Smart Grids - *IMDEA seminar series*, 30/05/2024.
- Machine Learning Inference in Programmable Switches with Random Forests - *BANYAN project summer school*, 26/05/2023.
- Practical Flow-Level Inference in Programmable Switches with Random Forests - *IMDEA Networks seminar series*, 03/05/2023.
- In-Network Machine Learning for Automatic Network Management - *UC3M Thesis Talk 2022*, 04/07/2022.
- An Introduction to In-Band Network Intelligence - *IMDEA Networks seminar series*, 03/11/2021.

COMMUNITY SERVICE

- Journal reviewer: IEEE/ACM Transactions on Networking (ToN)
- Journal reviewer: IEEE Transactions on Network and Service Management (TNSM)
- Journal reviewer: Elsevier Computer Networks (COMNET)
- Program committee member: ACM S3 2023 Workshop, PAM 2026, 2025; IEEE ICNP 2026; IEEE FINE 2026; ATC 2026
- Artifact evaluation committee member: SIGCOMM 2024
- Shadow program committee member: EuroSys 2024
- Publication chair: PAM 2023
- External reviewer: IEEE INFOCOM 2026, 2025, 2024; IEEE SECON 2024, 2023; IEEE GLOBECOM 2025, 2024; IFIP Networking 2024, 2023; ACM CIKM 2024; IEEE VNC 2023; WoWMoM 2022.

AWARDS

- Outstanding Doctoral Thesis Award (PhD), *UC3M*, Spain, December 2025
- Best Doctoral Thesis Award (PhD), *IEEE SPCOM Spain*, May 2025
- ACM SIGENERGY Student Travel Grant, *ACM e-Energy 2024*, April 2024
- IEEE ComSoC Student Travel Grant, *IEEE/IFIP NOMS 2024*, April 2024
- Best Demo Award, *IEEE NetSoft 2023*, June 2023

PROFESSIONAL MEMBERSHIPS

- Association for Computing Machinery (ACM), *Professional member*
- Institute of Electrical and Electronics Engineers (IEEE), *Member*
- IEEE Communications Society (ComSoc), *Member*

REPRESENTATIVE PUBLICATIONS

1. Aristide Tanyi-Jong Akem, Huiqi Yvonne Lu and Noa Zilberman, "In-Network Machine Learning for Real-Time Patient Monitoring on IoMT Edge Gateways," In: **IEEE Internet of Things Journal**, 2026. (Accepted) [\[PDF\]](#)
2. Aristide Tanyi-Jong Akem and Noa Zilberman, "HyNIC: hybrid in-network inference for line-rate anomaly detection on smartNICs," In: **2026 IEEE 12th International Conference on Network Softwarization (NetSoft)**, Berlin, Germany, 29 June - 3 July 2026. (Accepted) [\[PDF\]](#)
3. Aristide Tanyi-Jong Akem, Beyza Bütün, Michele Gucciardo and Marco Fiore, "Practical and General-Purpose Flow-Level Inference with Random Forests in Programmable Switches," In: **IEEE/ACM Transactions on Networking**, April

2025.[\[PDF\]](#)

4. Aristide Tanyi-Jong Akem, Beyza Bütün, Michele Gucciardo and Marco Fiore, “Jewel: Resource-Efficient Joint Packet and Flow Level Inference in Programmable Switches,” In: **IEEE INFOCOM 2024 - IEEE Conference on Computer Communications**, Vancouver, Canada, May 2024. [\[PDF\]](#)
5. Aristide Tanyi-Jong Akem, Michele Gucciardo and Marco Fiore, “Flowrest: Practical Flow-Level Inference in Programmable Switches with Random Forests,” In: **IEEE INFOCOM 2023 - IEEE Conference on Computer Communications**, New York Area, United States, May 2023. [\[PDF\]](#)

OTHER PUBLICATIONS

1. Aristide Tanyi-Jong Akem, Guillaume Fraysse and Marco Fiore, “Real-Time Encrypted Traffic Classification in Programmable Networks with P4 and Machine Learning,” In: **Int. J. Netw. Manag.**, vol 35, no 1, January 2025. [\[PDF\]](#)
2. Aristide Tanyi-Jong Akem, and Marco Fiore, “Towards Real-Time Intrusion Detection in P4-Programmable 5G User Plane Functions,” In: **The 7th European P4 Workshop (EuroP4’24), co-located with The 32nd IEEE International Conference on Network Protocols (ICNP)**, Charleroi, Belgium, October 2024. [\[PDF\]](#)
3. Aristide Tanyi-Jong Akem, Michele Gucciardo and Marco Fiore, “Ultra-Low Latency User-Plane Cyberattack Detection in SDN-based Smart Grids,” In: **In The 15th ACM International Conference on Future and Sustainable Energy Systems (E-Energy 24)**, Singapore, Singapore, June 2024. [\[PDF\]](#)
4. Aristide Tanyi-Jong Akem and Marco Fiore, “Towards Data-Driven Management of Mobile Networks through User Plane Inference,” In: **IEEE/IFIP Network Operations and Management Symposium**, Seoul, South Korea, May 2024. [\[PDF\]](#)
5. Aristide Tanyi-Jong Akem, Guillaume Fraysse and Marco Fiore, “Encrypted Traffic Classification at Line Rate in Programmable Switches with Machine Learning,” In: **NOMS 2024 - IEEE/IFIP Network Operations and Management Symposium**, Seoul, South Korea, May 2024. [\[PDF\]](#)
6. Michele Gucciardo, Beyza Bütün, Aristide Tanyi-Jong Akem and Marco Fiore, “Evaluating the Impact of Flow Length on the Performance of In-Switch Inference Solutions,” In: **IEEE INFOCOM 2024 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)**, Vancouver, Canada, May 2024. [\[PDF\]](#)
7. Aristide Tanyi-Jong Akem, Beyza Bütün, Michele Gucciardo and Marco Fiore, “Showcasing In-Switch Machine Learning Inference,” (Demo) In: **2023 IEEE 9th International Conference on Network Softwarization (NetSoft)**, Madrid, Spain, June 2023. [\[PDF\]](#) (Best Demo)
8. Beyza Bütün, Aristide Tanyi-Jong Akem, Michele Gucciardo and Marco Fiore, “Fast Detection of Cyberattacks on the Metaverse through User-plane Inference,” In: **2023 IEEE International Conference on Metaverse Computing, Networking and Applications (MetaCom)**, Kyoto, Japan, June 2023. [\[PDF\]](#)
9. Michele Gucciardo, Aristide Tanyi-Jong Akem, Beyza Bütün and Marco Fiore, “Demonstrating Flow-Level In-Switch Inference,” (Demo) In: **IEEE INFOCOM 2023 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)**, Hoboken, New Jersey, United States, May 2023. [\[PDF\]](#)
10. Aristide Tanyi-Jong Akem, Beyza Bütün, Michele Gucciardo, and Marco Fiore, “Henna: Hierarchical Machine Learning Inference in Programmable Switches,” In: **1st International Workshop on Native Network Intelligence (NativeNi’22)**, Rome, Italy, December 2022. [\[PDF\]](#)
11. Aristide Tanyi-Jong Akem and Edwin Mugume, “A Machine Learning Approach to Temporal Traffic-Aware Energy-Efficient Cellular Networks,” In: **2020 11th IEEE Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)**, New York, United States, October 2020. [\[PDF\]](#)

REFERENCES

Marco Fiore, Research Professor, IMDEA Networks Institute, Spain, marco.fiore@imdea.org

Noa Zilberman, Associate Professor, University of Oxford, United Kingdom, noa.zilberman@eng.ox.ac.uk

Guillaume Fraysse, Research Project Manager, Orange Innovation Networks, France, guillaume.fraysse@orange.com